

Heavy duty backpack



Note: Picture for Reference Only. It will be fully customized on the basis of products dimensions .

A heavy-duty **10L to 15L rugged daily carrier** is optimal. It protects against drops, prevents smaller accessories from rattling around, and isolates the power source safely.

To seamlessly transition between rugged outdoor fields and corporate or government offices, need a **hybrid "tactical-sleek" design**.

1. The Engineering Blueprint for The Layout

- **Suspended "Floating" Tablet Sleeve:** Do not let the tablet slot reach the very bottom of the bag. Require a **2cm suspended bottom gap** lined with high-density EVA foam. If the bag is dropped heavily on a rock, the biometric tablet will never make ground contact.
- **Segmented Battery Anchor Box:** Place the power bank pocket at the **bottom-center or mid-back** of the interior, independent of the tablet sleeve. It must feature a **heavy-duty elastic closure strap** to prevent a dense 20,000mAh mass from acting like a loose hammer inside the bag.
- **Dedicated Adapter Mesh Pocket:** Place a zippered, reinforced mesh pocket on the opposite wall (front flap) to store the charging brick and cables. This spreads the weight evenly and prevents metal charging prongs from scratching the tablet screen.
- **Neoprene Cable Organizers:** Request **3 to 4 elastic loop bands** right next to the adapter pocket to securely anchor loose biometric data cables and connection wires.

2. Engineering Directives

To ensure the bag transitions perfectly between outdoor field environments and professional offices, to apply these three structural rules :

- **Vertical Center-Mass Alignment (Anti-Sag):** To position the power bank pocket dead-center against the rear backplate, right below or adjacent to the tablet sleeve. This keeps the heaviest component tight against the wearer's spine, preventing the bag from pulling backward or sagging when worn.
- **Velcro Shielding Flap:** A dense layer separates the tablet compartment from the adapter and power bank compartments. This ensures that even under severe crushing forces during rough transit, the metal prongs or bricks cannot press into the tablet screen.
- **Clearance Tolerance:** To add a **tolerance buffer** to the tablet and power bank pocket dimensions. If a pocket is patterned exactly to the millimeter, internal stitching tension will make the pocket too tight to insert the hardware smoothly in the field.